Evaluation of Large Language Models as Emergency Department Revisit Predictors

Emma Chen[1,2†], Luyang Luo[1†], Fatma Gunturkun[3], Sraavya Sambara[1], Rushil Arora[3], Boyang Tom Jin[3], Pranav Rajpurkar[1*], David A Kim[3*]

[1]Harvard Medical School [2]Harvard SEAS [3]Stanford University

Motivation

Why Emergency Department (ED) Revisit Prediction?

- ~20% of discharged ED patients return within 30 days[1]
- Causes: incomplete evaluation, premature discharge, inadequate follow-up
- Challenge: multifactorial drivers across clinical, behavioral, and social factors

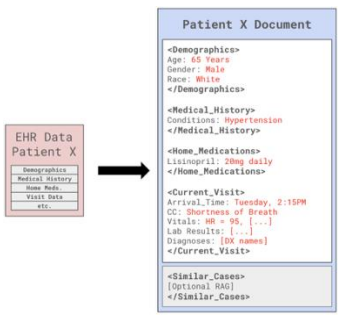
Why Large Language Models (LLMs) for This Task?

- Process unstructured clinical notes and radiology reports
- Leverage rich medical knowledge from pretraining
- Model longitudinal patient histories across visits

Contributions

1. **Comprehensive Benchmark** 138,010 ED visits evaluating multiple approaches to use LLMs for 30-day ED revisit prediction
2. **Open Challenge Identification** Current LLMs cannot exceed traditional structured-EHR-only machine learning approaches even with access to unstructured data
3. **Modeling Insights**
   1. Retrieval augmentation improves performance
   2. Embedding-based approaches outperform direct prediction
   3. LLMs' value may lie in representation learning, not end-to-end reasoning
4. **Reasoning Failure Analysis** Systematic examination of 17,488 reasoning chains reveals systematic biases

Methods



Dataset

- 138,010 discharged ED visits from Stanford ED (2/2021 - 2/2024)
- 87,112 unique patients
- 16.45% revisit rate in test set
- Structured + unstructured EHR data

Models Tested

Claude 3.7, GPT-4o, Deepseek R1, o3-mini, Voyage-3-Large, LLM2Vec[2] (Llama-3-8B-Instruct)

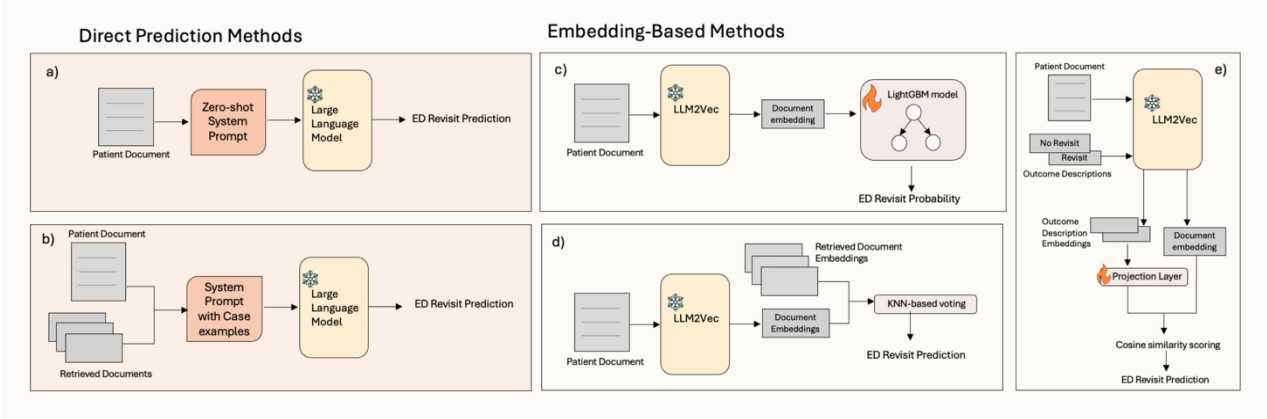
Approaches Evaluated

Direct Prediction Methods

Zero-shot: LLM generates risk assessment
RAG: Augmented with 10 similar cases

Embedding-Based Methods

Similarity Classification: Compare visit embeddings to outcome descriptions
LightGBM: Train classifier on embeddings
KNN: Predict based on similar case outcomes



Key Findings

Fig 1. Comparison of the best model across modeling paradigms

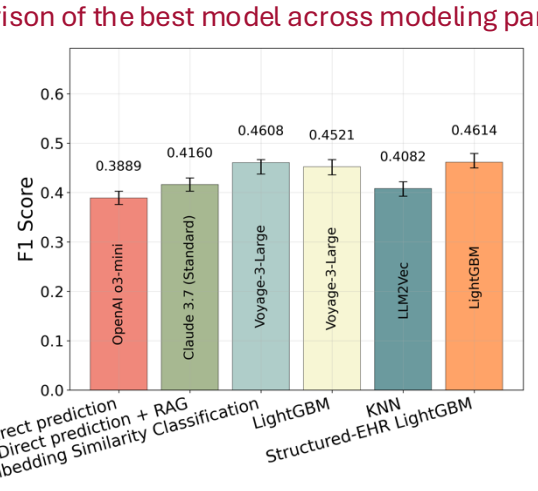


Fig 2. Performance of direct prediction with and without retrieval augmentation (RAG)

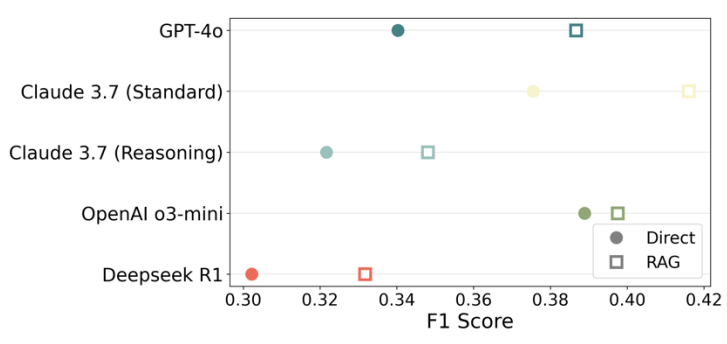


Fig 3. Qualitative analysis using GPT-4o on Claude 3.7's reasoning chains suggests possible ways reasoning changed predictions across 4,891 cases

Incorrectly changed low→high risk
88% overweighted medical histories, ignored protective factors
77% showed risk aversion bias
42% misinterpreted similar case statistics

Successfully identified missed high-risk cases
90% identified subtle social/demographic risks
63% caught untreated conditions
26% recognized escalating visit patterns

Model Performance (F1 Score)

Despite access to both structured AND unstructured data (radiology reports and longitudinal histories), LLM approaches did not outperform a simple LightGBM model using only structured features (Fig 1):

- Embedding Methods: 0.4505-0.4608
- Direct Prediction + RAG: 0.3867-0.4160
- Direct Prediction (zero-shot): 0.3022-0.3889
- *Baseline Structured EHR + LightGBM: 0.4614*

The Reasoning Paradox: When "Thinking" Hurts Performance

Claude 3.7's explicit reasoning capability unexpectedly degraded performance, with the standard mode (F1 = 0.3755) outperforming the reasoning mode (F1 = 0.3217) by 14% (Fig 2).

Automated qualitative analysis suggests that step-by-step reasoning may introduce systematic biases similar to known human cognitive errors in clinical decision-making[3] (Fig 3).

What Drives LLM Predictions? Top 5 Clinical Factors

Quantitative Analysis of 17,488 LLM Reasoning Chains:
Using k-means clustering on embedded risk factors, we identified 56 factor groupings. The most frequently considered align with clinical literature [4-6]:

- **Population Revisit Patterns** Comparisons to similar cases and hospital baseline rate
- **Very Recent Prior Visits** ED visits within days/weeks of current presentation
- **Frequent ED Utilization** Patterns of multiple recent visits
- **ED Length of Stay** Duration of encounters
- **Ongoing Treatment Protocols** Continued therapies requiring follow-up
- **Retrieval Validation:** 99.15% of predictions referenced the provided similar cases, confirming LLMs actively reason over retrieved examples.

[1] Dhaliwal, J. S., & Dang, A. K. (2024). Reducing hospital readmissions. In StatPearls [Internet]. StatPearls Publishing.
[2] BehnamGhader, P., Adlakha, V., Mosbach, M., Bahdanau, D., Chapados, N., & Reddy, S. (2024). Llm2vec: Large language models are secretly powerful text encoders. arXiv preprint arXiv:2404.05961.
[3] Sætrevik, B., Seeligmann, V. T., Frotvedt, T. F., & Keilegavlen Bondevik, Ø. (2024). Anchoring, confirmation and confidence bias among medical decision-makers. Collabra: Psychology, 10(1), 126223.
[4] Janda, S., & Sansuk, J. (2025). Factors Associated with Re-attendance at Emergency Departments Among Older Adults: A Cross-Sectional Analytical Study. INQUIRY: The Journal of Health Care Organization, Provision, and Financing, 62, 00469580251349652.
[5] Dufour, I., Chouinard, M. C., Dubuc, N., Beaudin, J., Lafontaine, S., & Hudon, C. (2019). Factors associated with frequent use of emergency-department services in a geriatric population: a systematic review. BMC geriatrics, 19(1), 185.
[6] Tangkulpanich, P., Yuksen, C., Kongchok, W., & Jenpanitpong, C. (2020). Clinical predictors of emergency department revisits within 48 hours of discharge; a case control study. Archives of academic emergency medicine, 9(1), e1.